

# GUI Team Planning Document

**Project:** Online Multiplayer Board Game

**Course:** SENG300 – Software Engineering

**Start Date:** Feb 26, 2025

**1st Meeting Date:** Feb 26, 2025

**2nd Meeting Date:** March 08, 2025

**3rd Meeting Date:** March 19, 2025

## 1. Overview

The **GUI team** is responsible for designing and implementing the **entire graphical user interface (GUI)** for the software using **JavaFX**. To enhance design and responsiveness, **Bootstrap may be used** where applicable.

The first phase involves **identifying and designing use cases**, as they serve as the foundation for the GUI. The **use case descriptions and diagrams** will be developed first. Use case descriptions will be collaboratively worked on in Google Docs, while diagrams will be created using Draw.io. Once finalized, wireframes will be developed using tools like Figma or Lucidchart, based on the identified use cases. To ensure quality, each team member will move their completed work to the review board in the issue tracker (Gitlab) for peer review.

After the wireframes are completed, the **GUI implementation phase** will begin, following the timeline outlined in **Table 2.0**.

GUI development will commence before Iteration 2 is due. The team will initialize the JavaFX project using the existing setup and begin implementation. Within **10-12 days after iteration 2**, a draft version of the GUI will be completed and tested to identify areas for improvement. Additionally, feedback from other teams after Iteration 2 will be incorporated for refinements. Any necessary improvements must be completed within **2 days** of the draft due date.

UI integration with the backend API will be done in parallel and must be completed **one day after all suggested improvements**. Once refinements are made, the final polished GUI will be prepared. A maximum of **2-3 days** is allocated for final testing and additional improvements. After successful testing, the **Integration Team** will take over **approximately 5 days to a week** before the final iteration is due.

## Handling Delays & Blockers

If any task is stuck, delayed, or requires urgent help, it must be communicated so that other team members can assist and ensure timely completion. The team will work collaboratively to resolve blockers and meet deadlines efficiently.

### GUI Team Responsibilities:

- **Use case diagrams and descriptions**
- **GUI mockups (wireframes)**
- **JavaFX-based GUI implementation**
- **UI testing & integration with backend**

## 2. Issue Board Utilization

We will use an **Issue Board (GitLab)** for task tracking.

- **GUI\_TO\_DO:** Tasks that are pending and need to be worked on.
- **GUI\_IN\_PROGRESS:** Tasks that are currently being worked on.
- **GUI\_REVIEW:** Completed tasks awaiting peer review.
- **GUI\_COMPLETED:** Tasks that have been reviewed, approved, and finalized.

**All team members must update the issue board after each work session.**

## 3. Work Schedule

Each team member will work 45 minutes per session, 3 days per week on assigned tasks.

Team meetings do not count towards work sessions!

## 4. Tools & Communication

- **Design:** Ms word, Figma
- **Development:** JavaFX, Scene Builder
- **Version Control:** GitLab (csgit.ucalgary.ca)
- **Task Management:** Issue Board (GitLab)
- **Communication:** WhatsApp

## **5. Key Deadlines & Milestones**

- **March 7** – GUI mockups & planning completed
- **March 28** – Basic UI layout implemented
- **April 1** – Backend integration & early testing completed
- **April 3** – Final GUI implementation ready & Hand over

## GUI Iteration 1 (Feb 27 - March 5): Planning & UI Design

**Table 1.0 (GUI Iteration 1)**

| <b>Task</b>           | <b>Responsible Members</b>                         | <b>Deadline</b> |
|-----------------------|----------------------------------------------------|-----------------|
| use case diagrams     | Member 1                                           | Feb 27          |
| Planning document     | Member 2                                           | Feb 27          |
| use case descriptions | Each Member 4 diagrams                             | March 3         |
| GUI mockups           | Each member (based on their use case descriptions) | March 5         |

Note: All tasks have been distributed and assigned to individual team members, with each member responsible for their respective tasks within the project.

### **Deliverables:**

Use Case Descriptions (**use\_case\_descriptions\_gui.pdf**)

Planning Document(**GUI\_timeline.pdf**)

Use Case Diagrams (**use\_case\_diagram\_gui.pdf**)

GUI Wireframe (**GUI\_wireframe.png**)

Project\_tracker (**task\_tracker\_GUI.pdf**)

## Use Cases Description

| Team Member | Assigned Use Cases                                                                                                                                                                             |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member 1    | <ol style="list-style-type: none"><li>1. Previous Page</li><li>2. Login</li><li>3. Sign Up</li><li>4. Forgot Password</li><li>5. Manage Profile</li></ol>                                      |
| Member 2    | <ol style="list-style-type: none"><li>1. Home Page</li><li>2. Logout</li><li>3. Add friend</li><li>4. Search Friend</li><li>5. Play Games</li></ol>                                            |
| Member 3    | <ol style="list-style-type: none"><li>1. Select Tic Tac Toe</li><li>2. Select Connect 4</li><li>3. Select Checkers</li><li>4. Choose Opponent</li><li>5. Friend (Opponent Selection)</li></ol> |
| Member 4    | <ol style="list-style-type: none"><li>1. In-Game Options</li><li>2. Game Chat</li><li>3. Top Players</li><li>4. Game History</li></ol>                                                         |
| Member 5    | <ol style="list-style-type: none"><li>1. Quit</li><li>2. Reset Password</li><li>3. Delete Profile</li><li>4. Edit Profile</li><li>5. Random (Opponent Selection)</li></ol>                     |

## GUI Iteration 2 (March 8 - March 28): Initial GUI Implementation

Table 2.0 (GUI iteration 2)

| Implement GUI                                                     | Deadline |
|-------------------------------------------------------------------|----------|
| Setup JavaFX project                                              | March 9  |
| Reusable components(Button, Modal, Forms, inputFields, SearchBar) | March 24 |
| Home Page                                                         | March 24 |
| Login/Signup Page                                                 | March 24 |
| Game Chat                                                         | March 28 |
| Game Screens                                                      | March 28 |
| Leaderboard                                                       | March 28 |
| Settings & Profile Pages                                          | March 28 |
| Draft GUI Submit                                                  | March 28 |

Note: All tasks have been distributed and assigned to individual team members, with each member responsible for their respective tasks within the project.

### Deliverables:

- Draft GUI

### GUI Iteration 3 (March 30 - April 3): Backend Integration & Testing

**Table 3.0 ( GUI Iteration 3)**

| Task                                  | Deadline |
|---------------------------------------|----------|
| Suggestions from Iteration 2 (if any) | March 30 |
| Connect UI to backend APIs            | March 31 |
| Run early UI testing                  | April 1  |
| UI Polish & Enhancements              | April 2  |
| Final Testing & Bug Fixes             | April 3  |
| Handover to Integration Team          | April 3  |

Note: All tasks have been distributed and assigned to individual team members, with each member responsible for their respective tasks within the project.

#### **Dependencies:**

- Final backend APIs must be ready

#### **Deliverables:**

- Fully Functional Static GUI (All screens)